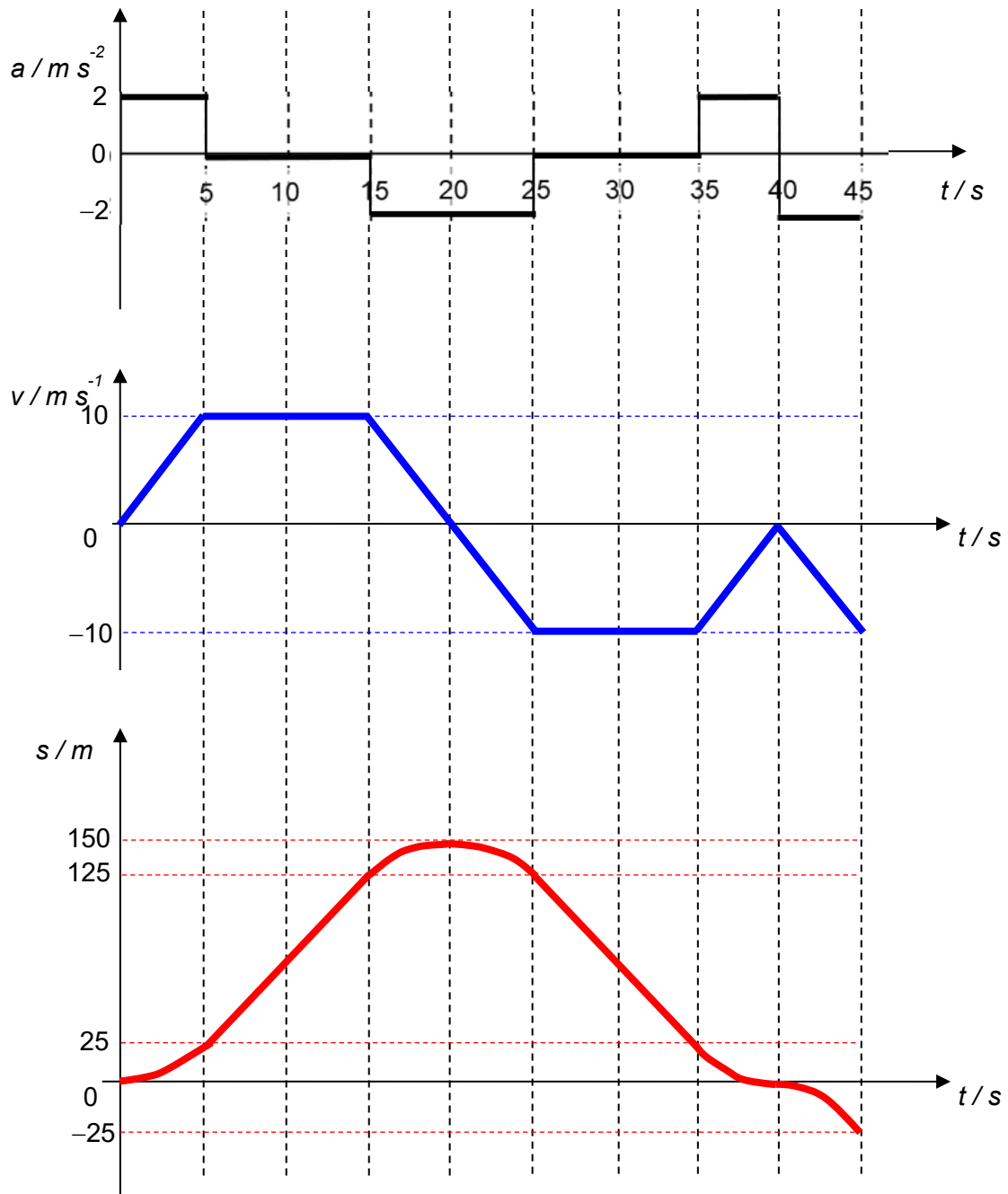
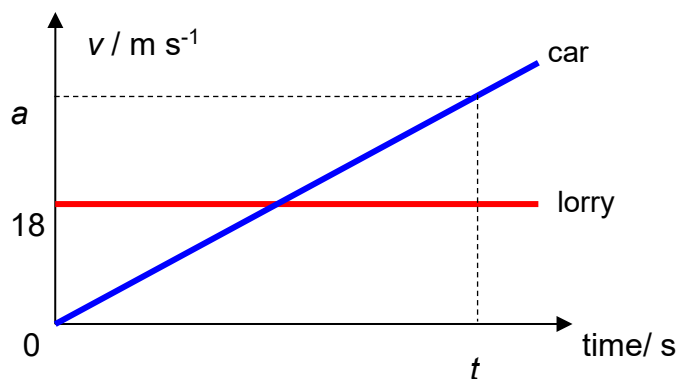


Tutorial 4 Motion and Forces Suggested Solutions

D1



Note: Displacement curve in this case must be smooth and without kinks!

D2

At time t ,

lorry's displacement from car's origin = car's displacement

$$v_{\text{lorry}} t + 60 = \frac{1}{2} a_{\text{car}} t^2$$

$$18 t + 60 = \frac{1}{2} (2.0) t^2$$

$$\frac{1}{2} (2.0) t^2 - 18 t - 60 = 0$$

$$t = \frac{18 \pm \sqrt{(-18)^2 - 4(1.0)(-60)}}{2.0}$$

$$= 20.9 \text{ s or } -2.9 \text{ s (reject)}$$

Distance travelled by car

$$s = \frac{1}{2} a_{\text{car}} t^2 = \frac{1}{2} (2.0) (20.9)^2 = 436 \text{ m}$$

D3 Displacement during the first 0.70 s

$$s_1 = ut = (13)(0.70) = 9.1 \text{ m}$$

Using $v^2 = u^2 + 2as$, distance travelled during braking

$$s_2 = \frac{v^2 - u^2}{2a} = \frac{0 - 13^2}{2(-4.5)} = 18.8 \text{ m}$$

Total distance travelled

$$s_{\text{total}} = s_1 + s_2 = 9.1 + 18.8 = 27.9 \text{ m } (> 25 \text{ m})$$

Distance travelled beyond the stop line

$$\Delta s = 27.9 - 25 = 2.9 \text{ m}$$

(Note: $s = ut + \frac{1}{2} at^2$ is not relevant in this case, $t \neq 0.70 \text{ s}$)

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- D4 (a)** Package moves upward briefly after release before falling to the ground
Taking upwards as positive and $s = -21$ m

$$s = ut + \frac{1}{2}at^2$$

$$-21 = 5t + \frac{1}{2}(-9.81)t^2$$

$$4.905t^2 - 5t - 21 = 0$$

$$t = 2.64 \text{ s or } -1.62 \text{ s (N.A.)}$$

- (b)** $v = u + at$
 $v = 5.0 + (-9.81)(2.64)$
 $v = -20.9 \text{ m s}^{-1} = 20.9 \text{ m s}^{-1} \text{ downwards}$

- D5 (a) (i)** Taking downwards as positive, $u = 0$

$$s = ut + \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2(1.00)}{9.79}} = 0.452 \text{ s (shown)}$$

- (ii)** Total time taken for the top edge of plate to reach the light beam

$$t_{\text{total}} = 0.052 + 0.452 = 0.504 \text{ s}$$

Let L be the length of metal plate, for top edge of plate

$$s = ut + \frac{1}{2}at^2$$

$$1.00 + L = 0 + \frac{1}{2}(9.79)(0.504)^2 = 1.24$$

$$L = 1.24 - 1.00 = 0.24 \text{ (2 sf)}$$

- (b)** Due to the presence of air resistance, the time taken for the plate to fall might have been longer than what was quoted in (a)(i) as this value was obtained under the assumption that air resistance is negligible.

The metal plate might have been released with a non-zero velocity.

- D6 (a) (i)** Using points $(124 \times 10^{-2}, 240.0 \times 10^{-3})$ and $(62 \times 10^{-2}, 110.0 \times 10^{-3})$,

$$\text{Gradient} = \frac{(240.0 - 110.0) \times 10^{-3}}{(124 - 62) \times 10^{-2}} = \frac{130.0 \times 10^{-3}}{62 \times 10^{-2}} = 0.210 \text{ s}^2 \text{ m}^{-1}$$

- (ii)** Taking downwards as positive,

$$s = ut + \frac{1}{2}at^2$$

Since $u = 0$, $t^2 = \left(\frac{2}{g}\right)s$

Hence, gradient of graph $= \frac{2}{g}$

$$g = \frac{2}{0.212} = 9.43 \text{ m s}^{-2}$$

(b) (i) 1. Plotting $\sqrt{s}$ against t

2. Plotting $\lg s$ against $\lg t$

(ii) $s = \frac{1}{2}gt^2$

$$\sqrt{s} = \sqrt{\frac{g}{2}}t$$

Plotting $\sqrt{s}$ against t would yield a straight line passing through the origin with gradient

$$\sqrt{\frac{g}{2}}$$

$$s = \frac{1}{2}gt^2$$

$$\lg s = \lg\left(\frac{g}{2}\right) + 2\lg t$$

Plotting $\lg s$ against $\lg t$ would yield a straight line with gradient 2 and

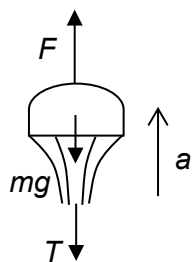
vertical intercept $\lg\left(\frac{g}{2}\right)$

D7 Due to the inertia of the bob of the plumbline, sketch should show the following features:

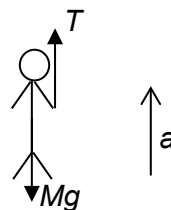
- (a) the plumbline will swing backward (negative x-direction) and remain inclined in that orientation.
- (b) the plumbline will swing forward (positive x-direction) and remain inclined in that orientation.
- (c) the plumbline will be vertical.
- (d) the plumbline will incline at a fixed angle, outward from the centre of the turn. This is due to an acceleration towards the centre of the turn as the car is constantly changing its direction of travel.

D8 (a) Let F be the force exerted on the parachute by the air and T be the force exerted by the parachute on the man.

Let m be the mass of the parachute while M is the mass of the man.



Forces on parachute



Forces on man

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Considering forces on the parachute

$$F - T - mg = ma \text{ ----- (1)}$$

Considering forces on the man

$$T - Mg = Ma \text{ ----- (2)}$$

Adding eqn (1) and eqn (2),

$$F - (M + m)g = (M + m)a \text{ ----- (3)}$$

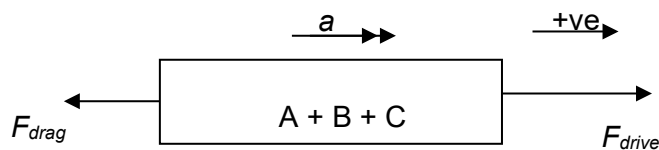
$$F = (M + m)(g + a) = (80 + 5.0)(9.81 + 2.4) = 1037 = 1040 \text{ N}$$

Alternatively, eqn (3) can be obtained by considering forces on the system of man and parachute and eqn (1) and eqn (2) can be dispensed with.

(b) From eqn (2)

$$T = M(g + a) = 80(9.81 + 2.4) = 977 \text{ N}$$

D9 (a) Considering A, B, C as the system:



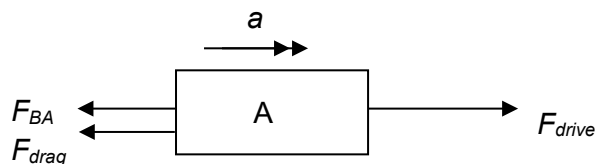
$$F_{net} = ma$$

$$F_{drive} - F_{drag} = ma$$

$$(2.0 \times 10^4 + 1.8 \times 10^4) - (9.0 \times 10^3 + 2 \times 6.0 \times 10^3) = [2(1.0 \times 10^4) + 8.0 \times 10^3] a$$

$$a = 0.61 \text{ m s}^{-2}$$

(b) Considering A as the system:



$$F_{net} = ma$$

$$F_{drive} - F_{drag} - F_{BA} = ma$$

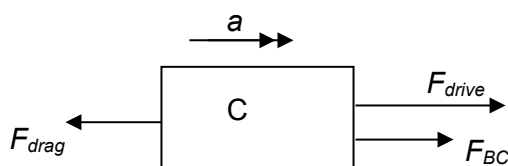
$$(2.0 \times 10^4) - (9.0 \times 10^3) - F_{BA} = (1.0 \times 10^4)(0.61)$$

$$F_{BA} = 4900 \text{ N}$$

By Newton's Third Law,

force that A exerts on B, $|F_{AB}|$ = force that B exerts on A, $|F_{BA}| = 4900 \text{ N}$.

(c) Considering C as the system:



$$F_{net} = ma$$

$$F_{drive} + F_{BC} - F_{drag} = ma$$

$$(1.8 \times 10^4) + F_{BC} - (6.0 \times 10^3) = (1.0 \times 10^4)(0.61)$$

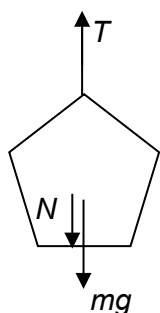
$$F_{BC} = -5900 \text{ N}$$

(F_{BC} is negative, i.e. pointing towards C because tow-bar is being compressed in this case.)

By Newton's Third Law,

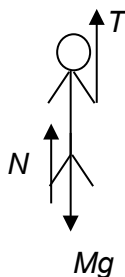
force that C exerts on B, $|F_{CB}|$ = force that B exerts on C, $|F_{BC}| = 5900 \text{ N}$.

D10



Forces on crate

↑ +ve



Forces on man

Take the upward direction to be positive.

Considering forces on the crate,

$$T - N - mg = ma \text{ ----- (1)}$$

Considering forces on the painter,

$$T + N - Mg = Ma \text{ ----- (2)}$$

eqn (2) – eqn (1),

$$2N - (M - m)g = (M - m)a$$

$$a = \frac{(2 \times 300) - (75 - 25) \times 9.81}{75 - 25} = 2.2 \text{ m s}^{-2} \text{ (upward)}$$

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- D11 (a) (i)** Reading increases / Reading is greater than W_2 .

As Bob lifts the dumbbell, he exerts an upward force larger than the weight of the dumbbell on the dumbbell to accelerate it upwards. (By N3L,) the dumbbell exerts an equal and opposite force (downwards) on him.

Hence the reading is greater than the total weight of Bob and dumbbell (W_2).

- (ii)** Reading is equal to W_1 .

Once the dumbbell leaves his hand, Bob no longer exerts any force on the dumbbell and vice versa.

Hence the reading is equal to the weight of Bob only.

- (b)** Yes, he can still see his image in the mirror.

When the man applies a force (or tension) on the rope, the mirror will experience the same force (or tension). Since the man and the mirror have the same mass, they will both move up at the same rate (or same acceleration).

Hence they will remain at the same level relative to each other.

- (c)** Scenario (1).

When F is applied, A, B and the bug will move with the same acceleration in both scenarios. In scenario (1), B has a smaller mass, hence it requires a smaller contact force to accelerate it. By Newton's 3rd Law, this results in a smaller contact force on the bug (assuming mass of bug is negligible).

- (d) (i)** Consider the 6.5 kg mass,

$$(6.5)(9.81) - T = 6.5a$$

$$63.765 - T = 6.5a \quad \dots (1)$$

Consider the 5.0 kg mass,

$$T - (5.0)(9.81) = 5.0a$$

$$T - 49.05 = 5.0a \quad \dots (2)$$

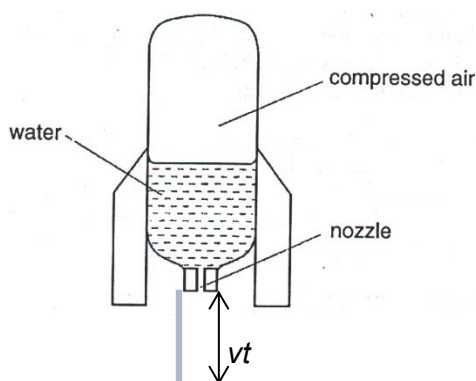
Solving,

$$a = 1.28 \text{ m s}^{-2}$$

$$T = 55.4 \text{ N}$$

- (ii)** 55.4 N

- D12 (a) (i)** In t seconds, the water ejected will move by a distance vt as shown:



The water traces out a cylindrical column of cross-sectional area $A = \pi r^2$ and length $L = vt$. Hence, the mass ejected in t seconds is $\rho AL = \rho \times \pi r^2 \times vt = \rho \pi r^2 vt$.

Mass of water ejected per second

$$\frac{dm}{dt} = \frac{\rho \pi r^2 vt}{t} = \rho \pi r^2 v$$

- (ii) Since $v_i = 0$, rate of change of momentum of the water

$$\frac{dp}{dt} = \Delta v \frac{dm}{dt} = \pi r^2 \rho v^2$$

- (b) Force on the water by the rocket = rate of change of momentum of the water

By Newton's 3rd Law of Motion,

$$|\text{upward thrust on rocket by water}| = |\text{force on the water by the rocket}|$$

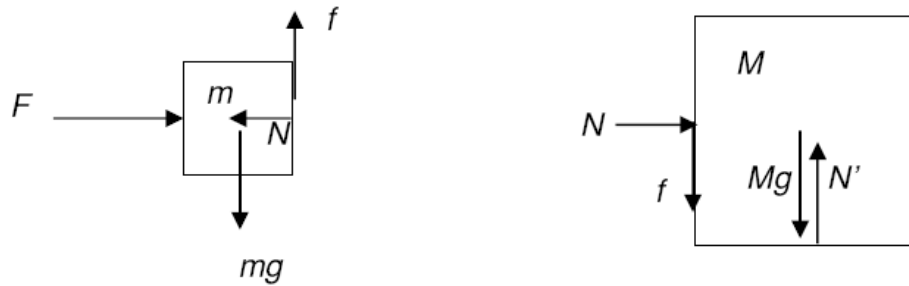
Hence, accelerating force or net force

$$\begin{aligned} F &= F_{\text{thrust}} - mg \\ &= \pi r^2 \rho v^2 - mg \end{aligned}$$

- (c) (i) Since the pressure difference between the compressed air and the air is unchanged, the body of water experiences the same pushing force downwards, thus the rate of change of momentum of the water forced through the nozzle is not affected. As a result, the initial thrust is unchanged.
- (ii) Since the volume of water is greater, the weight of the rocket is increased. The same initial thrust and larger weight will result in a smaller accelerating force F and a smaller acceleration a since $F = ma$.
- (iii) With a volume of 750 cm^3 of water, the volume of compressed air is smaller than before. When the smaller volume of air equalises with atmospheric pressure, its total volume after increase will also be smaller than before. Hence, not all the water will be pushed out and this increases the final mass of the rocket.

Challenging Questions Suggested Solutions

C1 (a)



(b) Consider vertical forces on m ,

$$mg \leq f_{\max} = \mu_s N \text{ ----- (1)}$$

Consider horizontal forces on M ,

$$N = Ma \text{ ----- (2)}$$

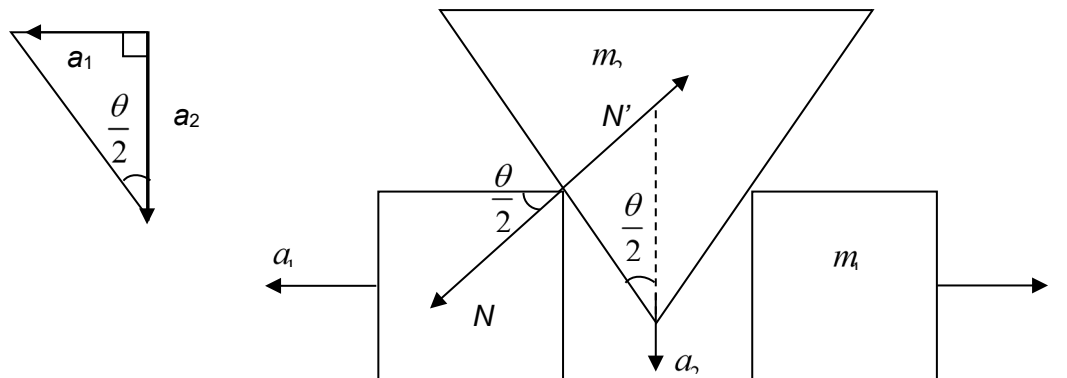
Combining equations (1) and (2),

$$a \geq \frac{mg}{\mu_s M} = \frac{1.6 \times 9.81}{0.38 \times 8.8} = 4.7 \text{ m s}^{-2}$$

Consider motion in horizontal direction for the system of the two blocks,

$$F = (M + m)a = (8.8 + 1.6)(4.7) = 49 \text{ N}$$

C2



$$\therefore a_1 = a_2 \tan \frac{\theta}{2}$$

$$N \cos \frac{\theta}{2} = m_1 a_1 = m_1 a_2 \tan \frac{\theta}{2}$$

By Newton's Third law, $N' = N$

Resolving for forces on triangular block,

$$m_2 g - m_2 a_2 = 2N' \sin \frac{\theta}{2} = 2 \left(\frac{m_1 a_2 \tan \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right) \sin \frac{\theta}{2} = 2m_1 a_2 \tan^2 \frac{\theta}{2}$$

$$m_2 g = a_2 \left(m_2 + 2m_1 \tan^2 \frac{\theta}{2} \right)$$

$$a_2 = \frac{m_2 g}{m_2 + 2m_1 \tan^2 \frac{\theta}{2}} \quad (\text{Ans})$$

$$a_1 = a_2 \tan \frac{\theta}{2} = \frac{m_2 g \tan \frac{\theta}{2}}{m_2 + 2m_1 \tan^2 \frac{\theta}{2}} \quad (\text{Ans})$$